JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI KRISHNAMOORTHI TRADERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

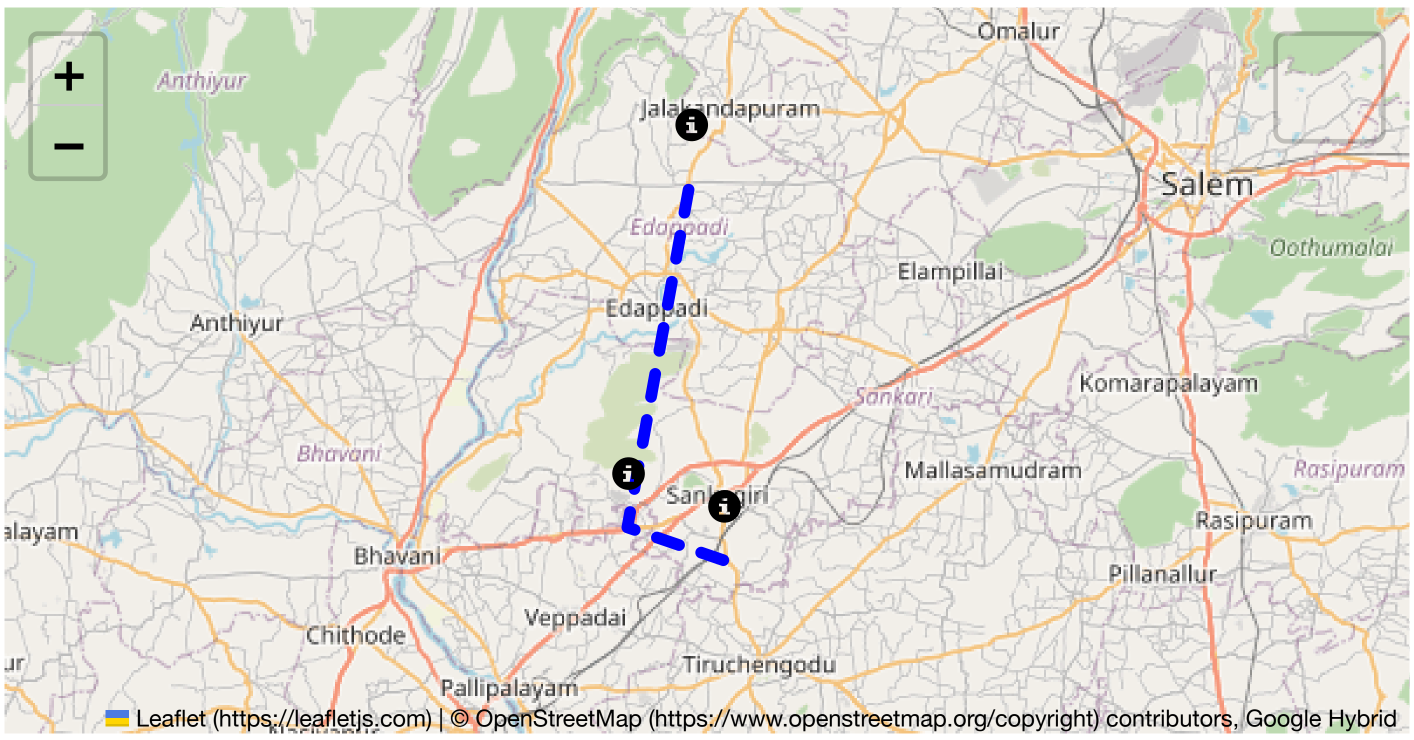
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

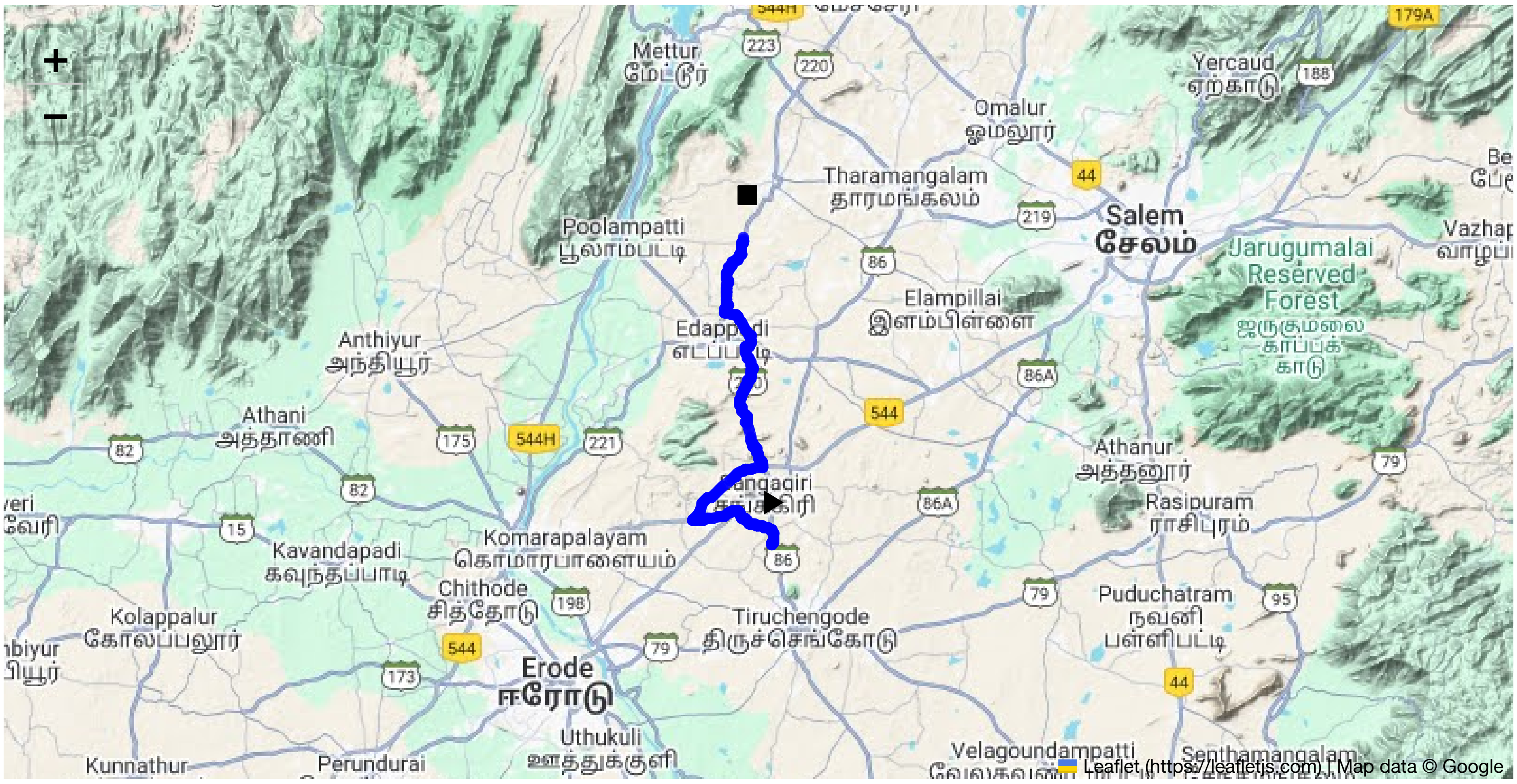
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 36.92 km
Estimated Duration: 0.9 hours

Adjusted Duration (Heavy Vehicle): 1.1 hours
Start: (11.4381, 77.8734)
End: (11.656981, 77.854179)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route starts at Sangagiri, Tamil Nadu, and proceeds towards Edappadi via Padaiveedu along Bhavani Main Road. The total distance is about 36.92 kilometers. Along the route, drivers will traverse a mix of highways and smaller local roads, primarily consisting of the NH 544 and connecting thoroughfares.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region experiences a hot, semi-arid climate. Key weather concerns include:

- **Summer Heat:** High temperatures can impact vehicle cooling systems and tire pressure.
- **Monsoon Rains (June-September):** Potential for localized flooding and reduced traction on roads.
- **Fog during the winter months (December-January):** Could impair visibility, particularly during early mornings.

3. Traffic Patterns

- **Peak Hours:** Typically between 8-10 AM and 5-8 PM, coinciding with work commutes.
- **Congestion-Prone Areas:** Near marketplaces and intersections in towns like Sangagiri and Edappadi. Traffic around schools and business districts can be significant during opening and closing hours.

4. Assessment of Road Quality and Infrastructure

Roads along the route generally maintain good quality, particularly the NH 544. However, some rural sections may have narrower lanes, potholes, and inadequate signage. Regular maintenance is needed for optimal safety.

5. Suggestions for Alternative Routes

In case of emergencies or roadblocks:

- Use the state highways intersecting NH 544 that bypass congested areas.
- Interior village roads offer potential detour options but require caution due to varying conditions.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permits Required:** Transport of hazardous materials requires specific permits from local authorities.
- **Time Restrictions:** Movement of heavy vehicles carrying hazardous materials may be restricted during peak hours and school zones.

7. Overview of Historical Incidents

While there are no widely reported recent incidents involving heavy vehicles on this route, neighboring areas have experienced episodes mainly due to road quality issues and driver fatigue.

8. Environmental Considerations and Sensitive Areas

- **Protected Areas:** Avoid ecological zones and farmlands, especially near water bodies, to prevent pollution.
- **Cultural Sites:** Maintain strict noise regulations and avoid disruptions near religious and cultural landmarks.

9. Analysis of Communication Coverage

- Communication networks are generally reliable along the NH 544.
- Potential dead zones are more likely in rural and less populated areas, particularly between Sangagiri and Padaiveedu.

10. Estimated Emergency Response Times

- **Urban Areas (Sangagiri and Edappadi):** Approximately 15-25 minutes.
- **Rural Segments:** Emergency response could take 30-45 minutes due to travel distance and road accessibility.

11. Overall Summary of Risk Assessment

The route presents moderate risk for heavy vehicle operations carrying hazardous materials. Key risks include weather-induced hazards during the monsoon, potential traffic congestion in urban areas, and varying road conditions on rural stretches. Preparedness involves confirming communication readiness, understanding local routes for detours, and adhering to transportation regulations. Regularly updated traffic and weather information can significantly reduce risks. Overall, cautious and informed driving should mitigate most potential issues.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60348, 77.83967	15 KM/Hr
2	Turn	High	11.43811, 77.87348	15 KM/Hr
3	Turn	Medium	11.43956, 77.87340	30 KM/Hr
4	Turn	High	11.43968, 77.87345	15 KM/Hr
5	Turn	High	11.44029, 77.87544	15 KM/Hr
6	Turn	Medium	11.45353, 77.85697	30 KM/Hr
7	Turn	Medium	11.45839, 77.85143	30 KM/Hr
8	Turn	Medium	11.46307, 77.84941	30 KM/Hr
9	Turn	Medium	11.46316, 77.84921	30 KM/Hr
10	Turn	High	11.45542, 77.81725	15 KM/Hr
11	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
12	Blind Spot	Blind Spot	11.49390, 77.86748	10 KM/Hr
13	Turn	Medium	11.52571, 77.85522	30 KM/Hr
14	Turn	Medium	11.55460, 77.85624	30 KM/Hr
15	Turn	High	11.57903, 77.85432	15 KM/Hr
16	Turn	Medium	11.58022, 77.85744	30 KM/Hr
17	Turn	Medium	11.60978, 77.84165	30 KM/Hr
18	Turn	Medium	11.64055, 77.84600	30 KM/Hr

Emergency Locations

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813

Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60348, 77.83967

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45353, 77.85697

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45839, 77.85143

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46307, 77.84941

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46316, 77.84921

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.45631, 77.81668

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49390, 77.86748

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.52571, 77.85522

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55460, 77.85624

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57903, 77.85432

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58022, 77.85744

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60978, 77.84165

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64055, 77.84600